

Iris ML Project - Complete Code Explanation (For Yagsha)

1. Project Goal

Train a Machine Learning model on the Iris dataset, save the model, make predictions, and run the project inside Docker.

2. What is the Iris Dataset?

The Iris dataset is a built-in dataset available in scikit-learn. It contains 150 flower records and 3 flower classes: Setosa, Versicolor, and Virginica. Each flower has 4 measurements: Sepal Length, Sepal Width, Petal Length, and Petal Width.

3. train.py Code

```
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
import pickle

iris = load_iris()

X = iris.data
y = iris.target

model = RandomForestClassifier()
model.fit(X, y)

with open("model.pkl", "wb") as f:
    pickle.dump(model, f)

print("Model trained and saved successfully!")
```

4. train.py Line-by-Line

load_iris() -> Loads the dataset.
iris.data -> Feature values (flower measurements).
iris.target -> Correct flower labels.
RandomForestClassifier() -> Creates ML model.
fit(X, y) -> Trains the model.
pickle.dump() -> Saves trained model.
model.pkl -> Saved trained model file.

5. app.py Code

```
import pickle

with open("model.pkl", "rb") as f:
    model = pickle.load(f)

sample = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(sample)

flowers = ["Setosa", "Versicolor", "Virginica"]

print("Prediction:", flowers[prediction[0]])
```

6. app.py Line-by-Line

pickle.load() -> Loads trained model.
sample -> New flower measurements.
model.predict() -> Predicts flower class.
prediction[0] -> Numeric result.
flowers list -> Converts number into flower name.

7. What Exactly Happened?

The dataset already existed in scikit-learn.
You did NOT manually enter 150 flower records.
The model learned from those 150 records during training.
Later you provided one new sample.
The model compared it with learned patterns and predicted Setosa.

8. Dockerfile

```
FROM python:3.11-slim

WORKDIR /app

COPY . .

RUN pip install scikit-learn

CMD ["python", "app.py"]
```

9. Dockerfile Explanation

FROM python:3.11-slim -> Base Python image.
WORKDIR /app -> Working directory.
COPY . . -> Copies project files.
RUN pip install scikit-learn -> Installs ML dependency.
CMD -> Runs app.py when container starts.

10. Docker Commands Used

```
docker build -t iris-ml .
```

Creates image.

```
docker run iris-ml
```

Runs container.

```
docker images
```

Lists images.

`docker ps -a`
Lists containers.

11. Interview Questions

Q. What is Machine Learning?

A. Teaching computers to learn patterns from data.

Q. What is Random Forest?

A. A machine learning algorithm based on multiple decision trees.

Q. What is `model.pkl`?

A. Saved trained model file.

Q. Why Docker?

A. To package code, model, Python, and dependencies together.

Q. What is the difference between Image and Container?

A. Image is a blueprint; Container is a running instance.

Q. Did you create the dataset?

A. No. The Iris dataset is provided by scikit-learn. I trained a model on it and containerized the application.

12. One-Line Project Explanation

I trained a Random Forest model on the Iris dataset, saved it using Pickle, and containerized the application using Docker for consistent deployment.